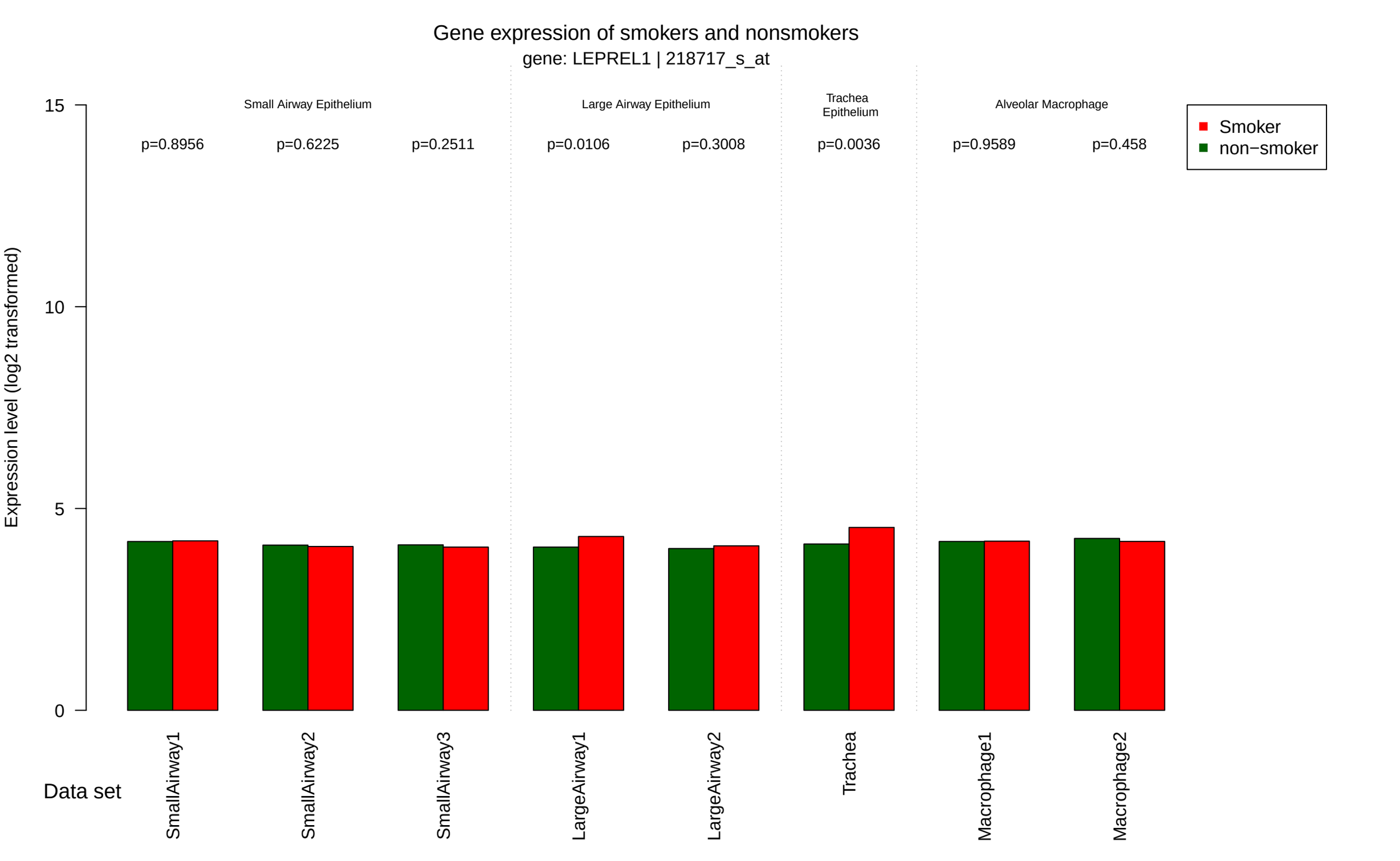
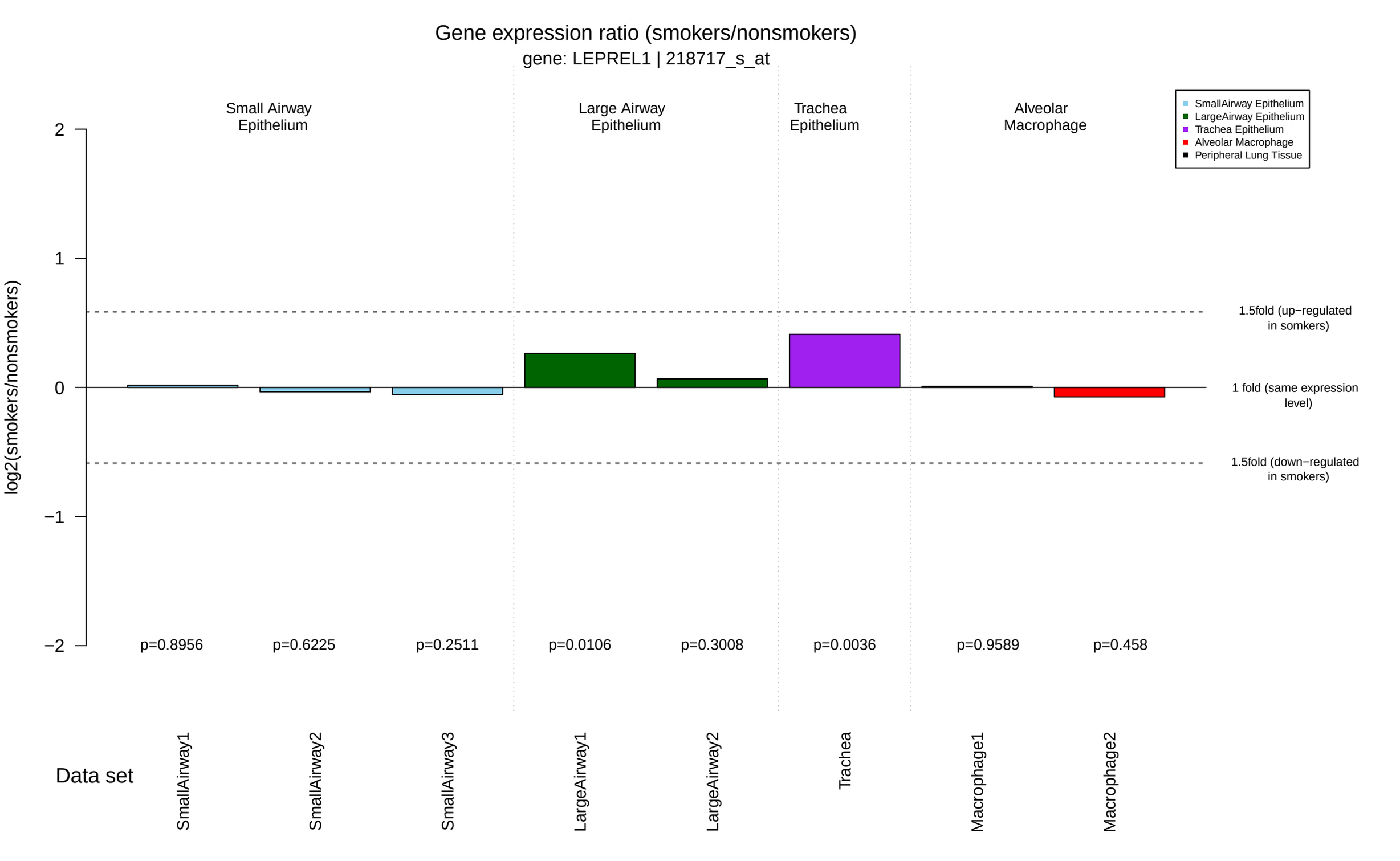
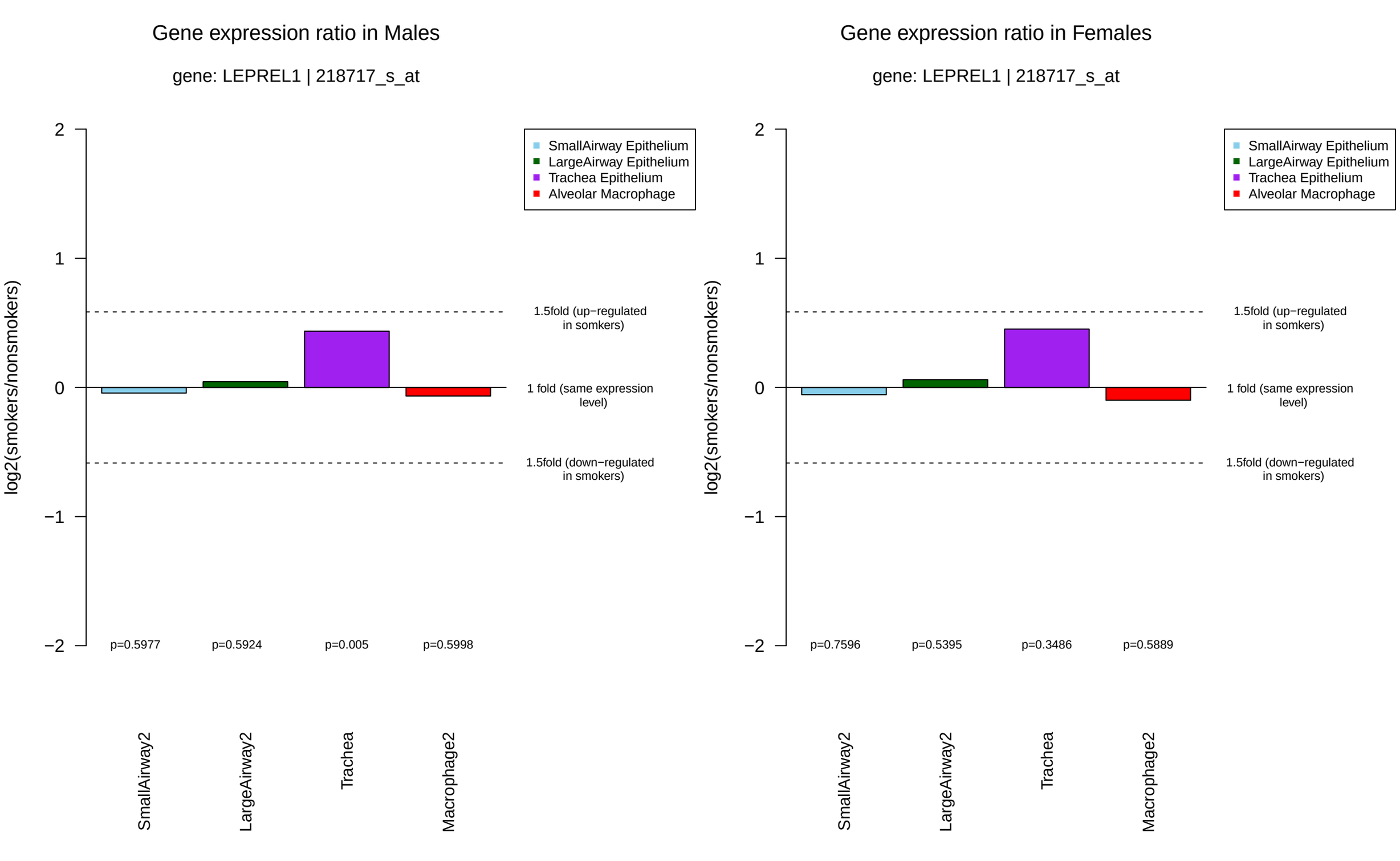
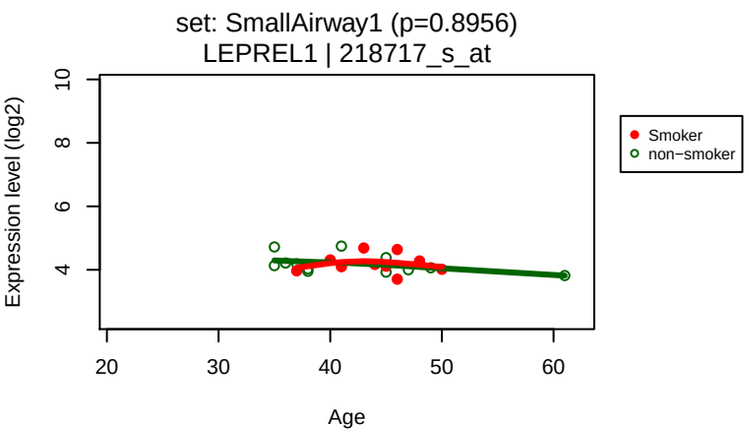
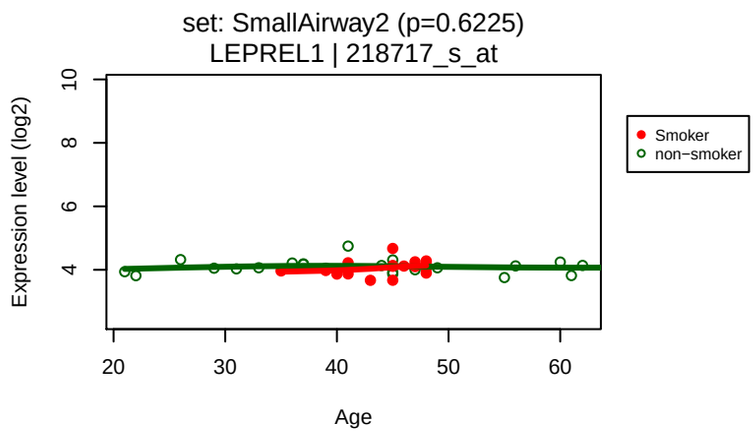
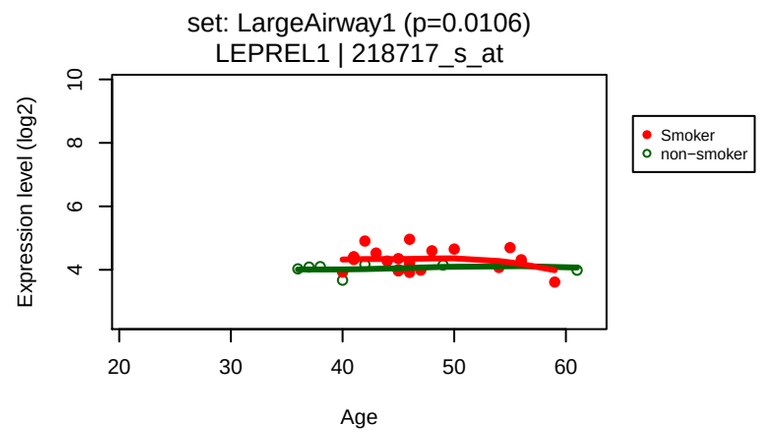

Gene expression of smokers and nonsmokers
gene: LEPREL1 | 218717_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8956
p=0.6225
p=0.2511
p=0.0106
p=0.3008
p=0.0036
p=0.9589
p=0.458
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

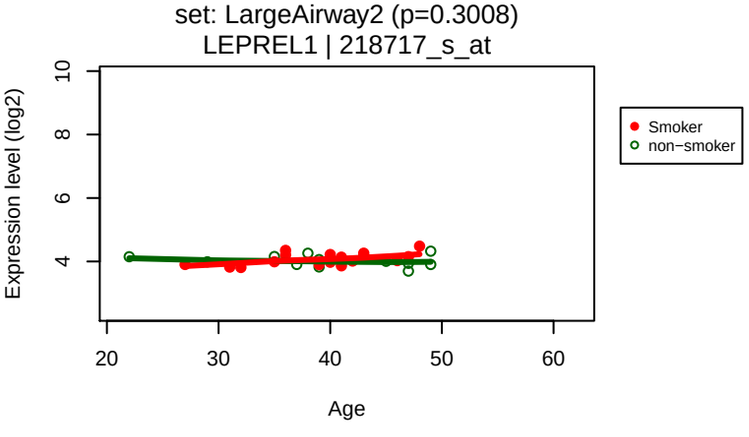

Gene expression ratio (smokers/nonsmokers)
gene: LEPREL1 | 218717_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8956
p=0.6225
p=0.2511
p=0.0106
p=0.3008
p=0.0036
p=0.9589
p=0.458
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

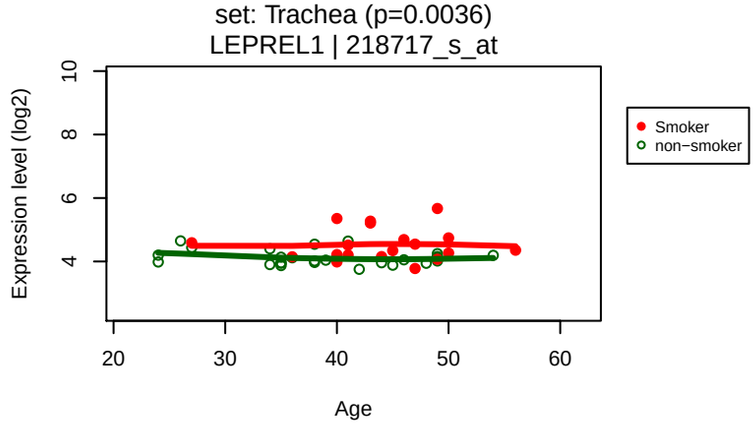

Gene expression ratio in Males
gene: LEPREL1 | 218717_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5977
p=0.5924
p=0.005
p=0.5998
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LEPREL1 | 218717_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7596
p=0.5395
p=0.3486
p=0.5889
SmallAirway2
LargeAirway2
Trachea
Macrophage2

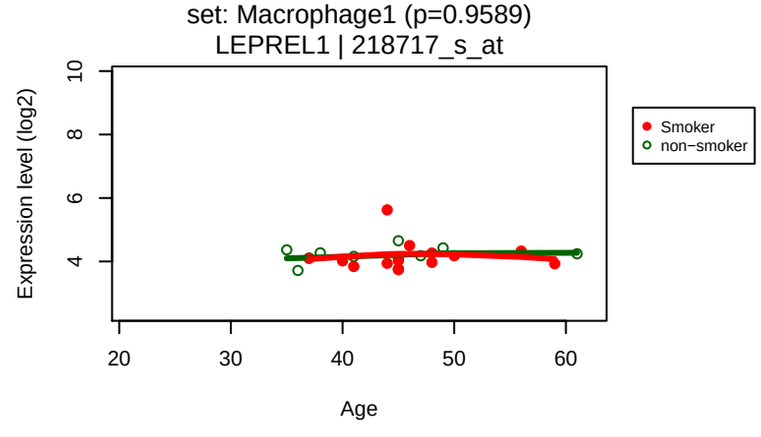

set: SmallAirway1 (p=0.8956)
LEPREL1 | 218717_s_at
Expression level (log2)
Age
Smoker
non−smoker

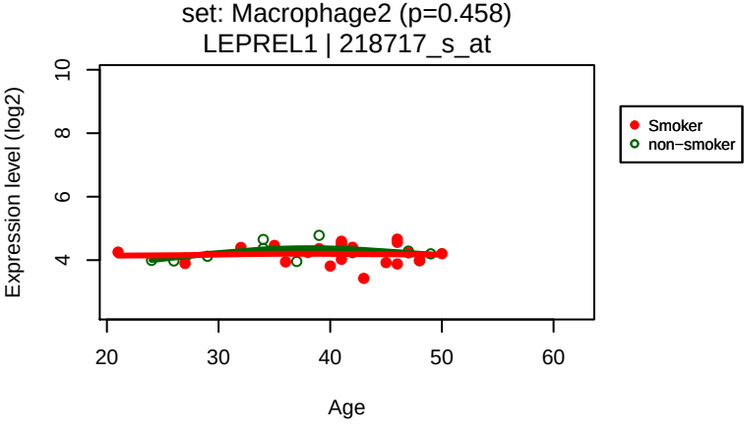

set: SmallAirway2 (p=0.6225)
LEPREL1 | 218717_s_at
Expression level (log2)
Age
Smoker
non−smoker

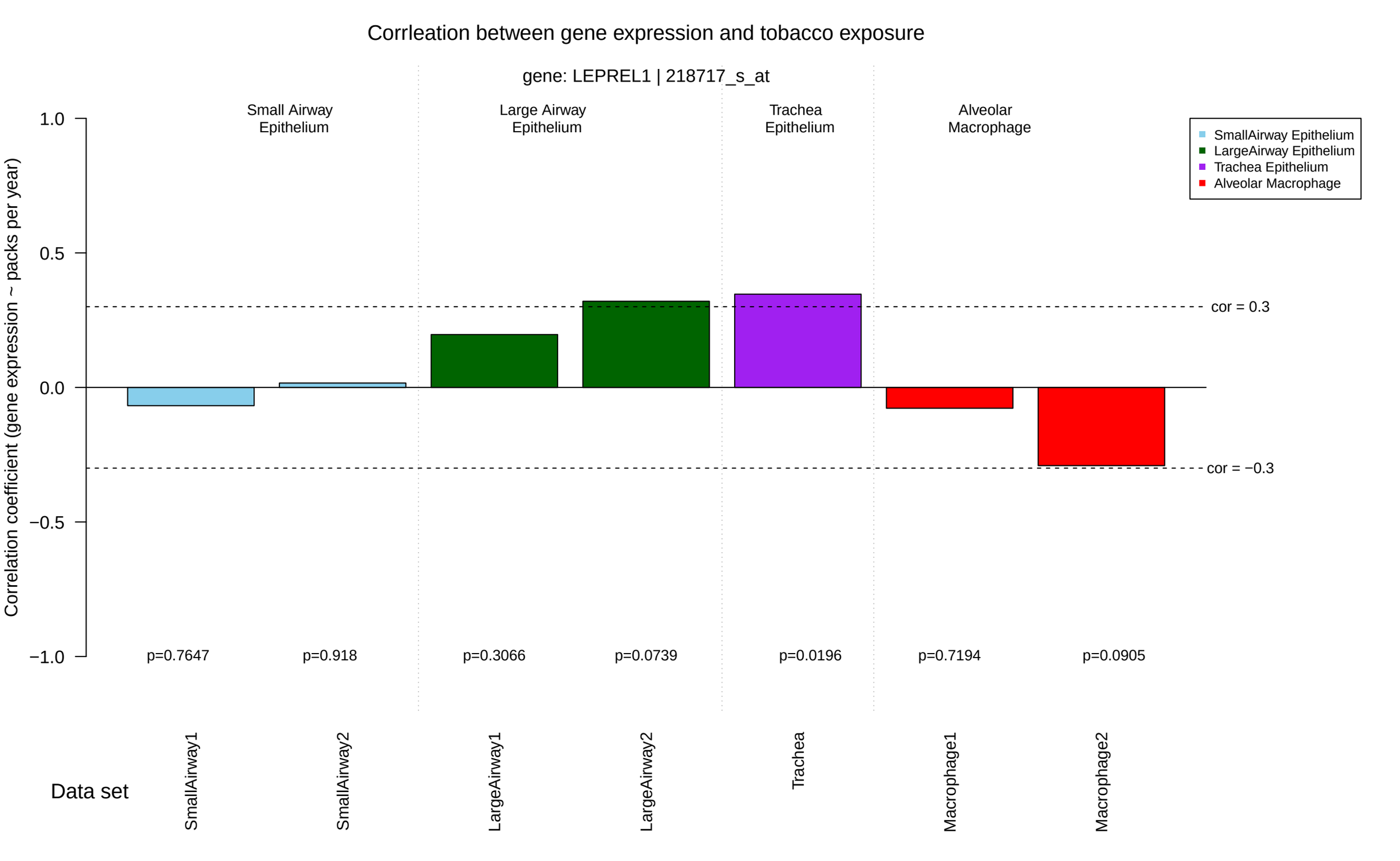

set: LargeAirway1 (p=0.0106)
LEPREL1 | 218717_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3008)
LEPREL1 | 218717_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0036)
LEPREL1 | 218717_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9589)
LEPREL1 | 218717_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.458)
LEPREL1 | 218717_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LEPREL1 | 218717_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7647
p=0.918
p=0.3066
p=0.0739
p=0.0196
p=0.7194
p=0.0905
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2